

**DEPARTMENT OF MECHANICAL ENGINEERING
FACULTY OF ENGINEERING AND ARCHITECTURE
MBM UNIVERSITY JODHPUR**

Subject: 5ME42A(ME) – Manufacturing Technology

Assignment # 2

Topic: Metal Working

Note: Make suitable assumptions. Use neat sketches to represent essential terms

1. Derive expression for:
 - a) Roll separating force, driving torque and power in rolling operation.
 - b) Maximum force required for forging.
 - c) Drawing force, power and maximum allowable reduction in drawing operation.
 - d) Drawing force, power and minimum possible radius of cup in deep drawing operation.
 - e) Maximum bending force and spring back in bending operation.
 - f) Frictional power loss and maximum force required in extrusion process.
2. Write about advantages and disadvantages of hot and cold working.
3. What are the effects of friction in metal working process?
4. State about importance of lubrication in metal working processes.